

Array Assignment

1. create a class called **ComplexNumber** having 2 data members as **number1** and **number2**. Create a pair of **getter** and **setter** for the above. Create a method called **computeComplexNumber** that will multiply the values of the 2 data members and return it in the form of **int**. For the above create an array of **ComplexNumber** having 5 elements and in a loop ask the user to enter **number1** and **number2** 5 times each such that each object of the **ComplexNumber** hold appropriate values. Finally call the **computeComplexNumber** on each object of the array to display the result.

```
import java.util.Scanner;

public class ComplexNumber {

    private int number1;
    private int number2;

    public void setNumber1(int number1) {
        this.number1 = number1;
    }

    public int getNumber1() {
        return number1;
    }

    public void setNumber2(int number2) {
        this.number2 = number2;
    }

    public int getNumber2() {
        return number2;
    }

    public int computeComplexNumber() {
        return number1 * number2;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        ComplexNumber arr[] = new ComplexNumber[5];
```

```

        for (int i = 0; i < 3; i++) {

            arr[i] = new ComplexNumber();

            System.out.println("Enter Number 1 : ");
            int n1 = sc.nextInt();

            System.out.println("Enter Number 2 : ");
            int n2 = sc.nextInt();

            arr[i].setNumber1(n1);
            arr[i].setNumber2(n2);
        }

        System.out.println("Result : ");

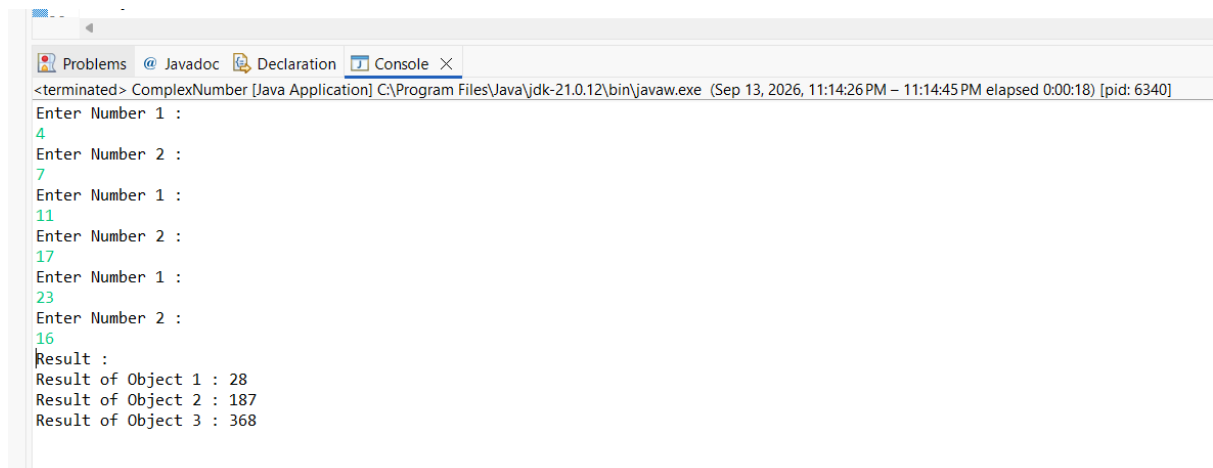
        for (int i = 0; i < 3; i++) {

            int result = arr[i].computeComplexNumber();

            System.out.println("Result of Object " + (i + 1) + " : " + result);
        }

        sc.close();
    }
}

```



```

Problems @ Javadoc Declaration Console X
<terminated> ComplexNumber [Java Application] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 13, 2026, 11:14:26 PM – 11:14:45 PM elapsed 0:00:18) [pid: 6340]
Enter Number 1 :
4
Enter Number 2 :
7
Enter Number 1 :
11
Enter Number 2 :
17
Enter Number 1 :
23
Enter Number 2 :
16
Result :
Result of Object 1 : 28
Result of Object 2 : 187
Result of Object 3 : 368

```

2. Write a Java program to sort an numeric array. The size of the array will be taken from the user, he specifies the size all the elements of the array will be taken as input and the array will be sorted.

```

public class Exe1SortNumArray {

    public static void main(String[] args)
    {
        System.out.println("Enter the size of array : ");
        int size = ConsoleInput.getInt();

        int arr[] = new int[size];

        System.out.println("Enter the elements : ");

        for (int i = 0; i < size; i++)
        {
            arr[i] = ConsoleInput.getInt();
        }

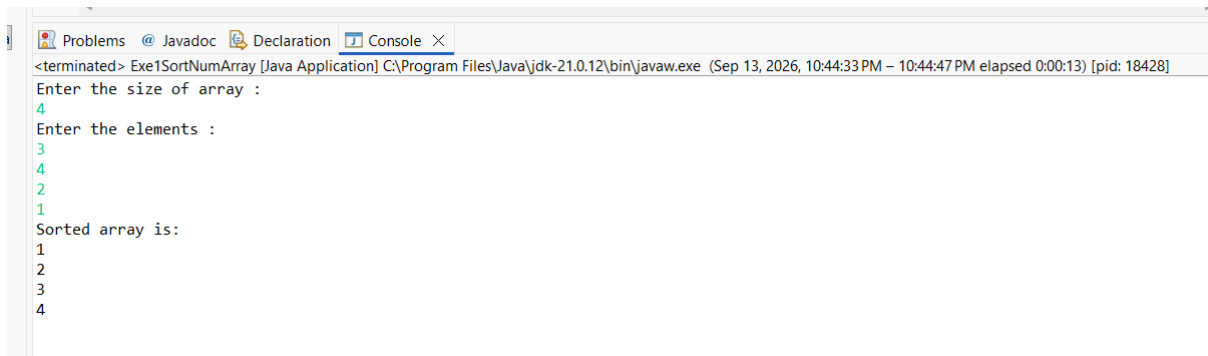
        for (int i = 0; i < size - 1; i++)
        {
            for (int j = i + 1; j < size; j++)
            {
                if (arr[i] > arr[j])
                {
                    int temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }

        System.out.println("Sorted array is:");

        for (int i = 0; i < size; i++)
        {
            System.out.println(arr[i]);
        }
    }
}

```

}



The screenshot shows a Java IDE window with a console tab. The console output is as follows:

```
<terminated> Exe1SortNumArray [Java Application] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 13, 2026, 10:44:33 PM - 10:44:47 PM elapsed 0:00:13) [pid: 18428]
Enter the size of array :
4
Enter the elements :
3
4
2
1
Sorted array is:
1
2
3
4
```

3. Modify the exercise and Write a Java program to sum values of an array

```
public class SumOfArray {

    public static void main(String[] args) {
        // TODO Auto-generated method stub

        System.out.println("Enter the Size of the Array : ");
        int size = ConsoleInput.getInt();

        int arr[] = new int[size];

        System.out.println("Enter the Array Elements : ");

        for (int i = 0; i < size; i++)
        {
            arr[i] = ConsoleInput.getInt();
        }

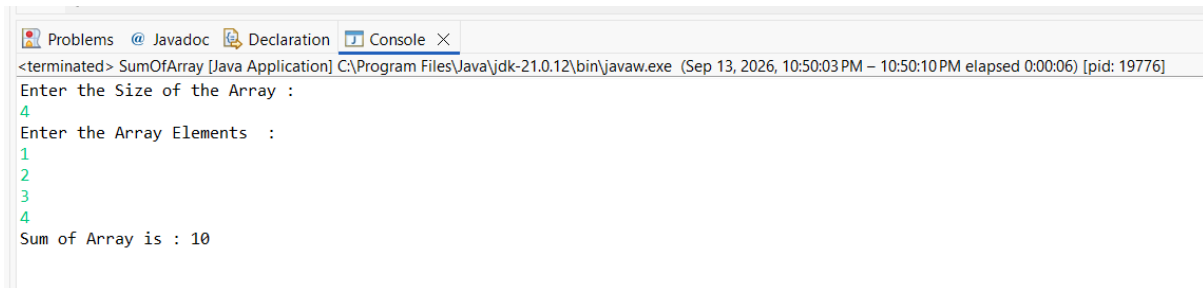
        int sum = 0;

        for(int i = 0; i < size; i++)
        {
            sum = sum + arr[i];
        }

        System.out.println("Sum of Array is : " + sum);

    }
}
```

}



```
<terminated> SumOfArray [Java Application] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 13, 2026, 10:50:03 PM - 10:50:10 PM elapsed 0:00:06) [pid: 19776]
Enter the Size of the Array :
4
Enter the Array Elements :
1
2
3
4
Sum of Array is : 10
```

4. Modify exercise Write a Java program to calculate average value of an array element

```
public class AvgValueOfArray {

    public static void main(String[] args) {

        System.out.println("Enter the Size of the Array : ");
        int size = ConsoleInput.getInt();

        int arr[] = new int[size];

        System.out.println("Enter the Array Elements : ");
        for(int i = 0; i < size; i++)
        {
            arr[i] = ConsoleInput.getInt();
        }

        int sum = 0;

        for(int i = 0; i < size; i++)
        {
            sum = sum + arr[i];
        }
        System.out.println("Sum of the Array Elements : " + sum);

        double Average = (double) sum / size;

        System.out.println("Average of the Array Elements : " + Average);
    }
}
```

```

    }

}

```

```

Problems  @ Javadoc  Declaration  Console  X
<terminated> AvgValueOfArray [Java Application] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 13, 2026, 10:52:32 PM – 10:52:43 PM elapsed 0:00:10) [pid: 10416]
Enter the Size of the Array :
4
Enter the Array Elements :
1
2
3
4
Sum of the Array Elements : 10
Average of the Array Elements : 2.5

```

5. Modify exercise Write a Java program to copy an array by iterating the array

```

public class ArrayCopyIterator {

    public static void main(String[] args) {

        System.out.println("Enter the Size of the Array : ");
        int size = ConsoleInput.getInt();
        int arr[] = new int[size];
        System.out.println("Enter the Array Elements : ");
        for (int i = 0; i < size; i++)
        {
            arr[i] = ConsoleInput.getInt();
        }

        int copy[] = new int[size];
        for (int i = 0; i < size; i++)
        {
            copy[i] = arr[i];
        }
        System.out.println("Original Array : ");
        for (int i = 0; i < size; i++)
        {
            System.out.print(arr[i] + " ");
        }
        System.out.println("\nCopied Array : ");
        for (int i = 0; i < size; i++)
        {
            System.out.print(copy[i] + " ");
        }
    }
}

```

```
}  
  
}  
  
}
```



```
<terminated> ArrayCopyIterator [Java Application] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 13, 2026, 10:56:10 PM - 10:56:20 PM elapsed 0:00:09) [pid: 8624]  
Enter the Size of the Array :  
4  
Enter the Array Elements :  
7  
4  
10  
11  
Original Array :  
7 4 10 11  
Copied Array :  
7 4 10 11
```

6. Modify exercise 2 Write a Java program to find the maximum and minimum value of an array.

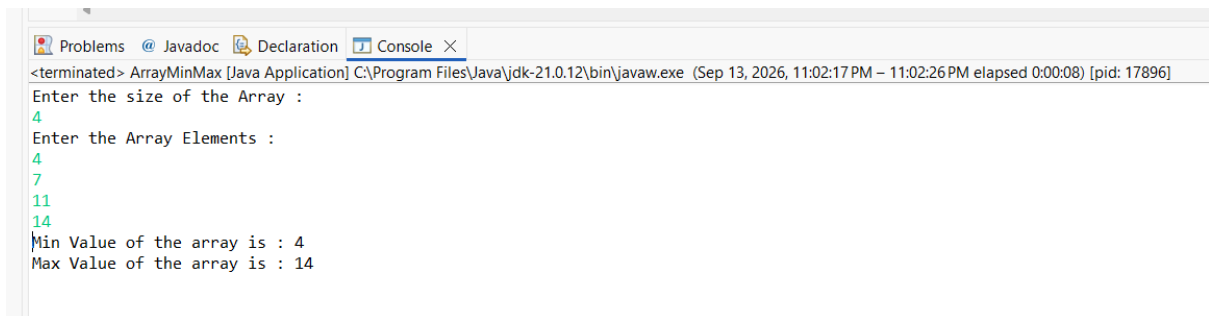
```
public class ArrayMinMax {  
  
    public static void main(String[] args) {  
  
        System.out.println("Enter the size of the Array : ");  
        int size = ConsoleInput.getInt();  
  
        int arr[] = new int[size];  
  
        System.out.println("Enter the Array Elements : ");  
  
        for(int i = 0; i < size; i++)  
        {  
            arr[i] = ConsoleInput.getInt();  
  
        }  
  
        int max = arr[0];  
        int min = arr[0];  
  
        for (int i = 1; i < size; i++)  
        {  
            if (arr[i] < min)  
            {  
                min = arr[i];  
            }  
        }  
    }  
}
```

```

        if (arr[i] > max)
        {
            max = arr[i];
        }
    }

    System.out.println("Min Value of the array is : " + min);
    System.out.println("Max Value of the array is : " + max);
}
}

```



```

Problems @ Javadoc Declaration Console X
<terminated> ArrayMinMax [Java Application] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 13, 2026, 11:02:17 PM - 11:02:26 PM elapsed 0:00:08) [pid: 17896]
Enter the size of the Array :
4
Enter the Array Elements :
4
7
11
14
Min Value of the array is : 4
Max Value of the array is : 14

```

7. Modify exercise Write a Java program to reverse an array of integer values

```

public class RevArray {

    public static void main(String[] args) {

        System.out.println("Enter the Size of an Array : ");
        int size = ConsoleInput.getInt();

        System.out.println("Enter the Elements of an Array : ");

        int arr[] = new int[size];

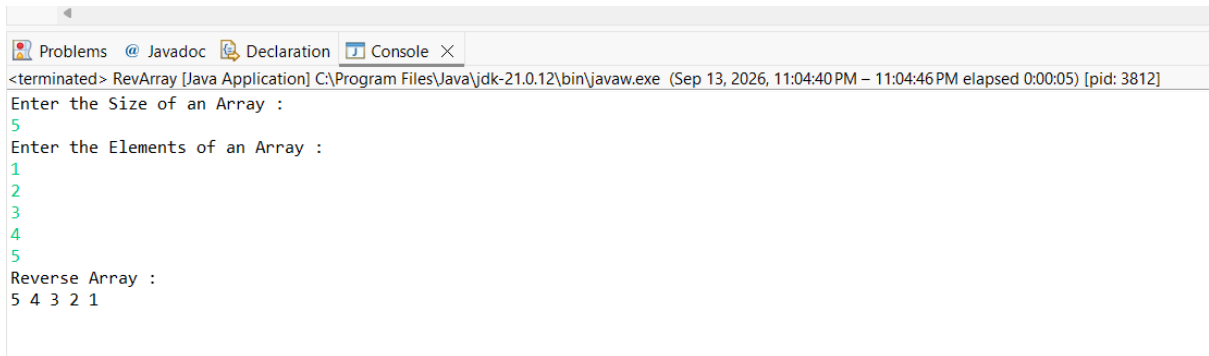
        for (int i = 0; i < size; i++)
        {
            arr[i] = ConsoleInput.getInt();
        }

        System.out.println("Reverse Array : ");
        for (int i = size - 1; i >= 0; i--)
        {
            System.out.print(arr[i] + " ");
        }
    }
}

```

```
}
```

```
}
```



```
Problems @ Javadoc Declaration Console X
<terminated> RevArray [Java Application] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 13, 2026, 11:04:40 PM – 11:04:46 PM elapsed 0:00:05) [pid: 3812]
Enter the Size of an Array :
5
Enter the Elements of an Array :
1
2
3
4
5
Reverse Array :
5 4 3 2 1
```

8. Modify exercise 2 Write a Java program to find the duplicate values of an array of integer values

```
public class DuplicateArr {

    public static void main(String[] args) {

        System.out.println("Enter the Size of an Array : ");

        int size = ConsoleInput.getInt();


        System.out.println("Enter the Elements of an Array : ");


        int arr[] = new int[size];


        for (int i = 0; i < size; i++)

        {

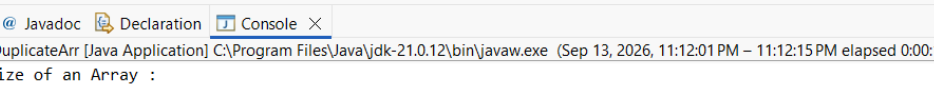
            arr[i] = ConsoleInput.getInt();

        }


        System.out.println("Duplicate Values : ");

        for (int i = 0; i < size; i++)

        {
```



The screenshot shows the 'Console' tab of a Java IDE. The console output is as follows:

```
<terminated> DuplicateArr [Java Application] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 13, 2026, 11:12:01 PM – 11:12:15 PM elapsed 0:00:13) [pid: 3924]
Enter the Size of an Array :
5
Enter the Elements of an Array :
1
2
4
4
7
Duplicate Values :
4
```

9. **Modify exercise 2 to accept 2 different values in 2 different arrays and find the common elements between two arrays**

```
public class CommEle2Arr {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Enter the Size of Array 1 : ");
        int size1 = ConsoleInput.getInt();

        int arr1[] = new int[size1];

        System.out.println("Enter the Elements of Array 1 : ");

        for (int i = 0; i < size1; i++) {
            arr1[i] = ConsoleInput.getInt();
        }
    }
}
```

